

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.		(ii)		1 mol of H_2SiF_6 (144g) gives 6 mol of F^- ions ($6 \times 19\text{g}$) = 114g (1) $\therefore$ 114 mg of fluoride ions from 144 mg H_2SiF_6 The increase in fluoride ion concentration needed is $0.76 - 0.15 = 0.61 \text{ mg dm}^{-3}$ (1) Amount of H_2SiF_6 needed is $\frac{144 \times 0.61}{114} = 0.77 \text{ mg}$ (1) Accept alternative method Award (3) for cao			3	3	2	
	(e)			MgSiF_6 M_r 166 $\therefore$ Moles of $\text{MgSiF}_6 = 2.6/166 = 1.566 \times 10^{-2}$ (1) $[\text{H}^+]$ is $\therefore 4 \times 1.566 \times 10^{-2} = 0.06265 \text{ mol dm}^{-3}$ (1) $\text{pH} = -\log_{10} [\text{H}^+] = 1.20$ (1) ecf possible		3		3	2	
Question 10 total					3	9	5	17	8	0